

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems

COURSE CODE: CSA0508

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Concept Mapping (Medium)

Assessment Tool Weightage: 35%

Dr G. AYYAPPAN

QUESTIONS		Total Marks: 50	
Q.No	Question	Bloom's Level	Marks
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand	5
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand	5
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand	5
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand	5
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand	5
6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand	5
7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand	5

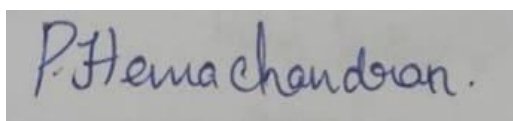
8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand	5
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand	5
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.	BL1 – Remember	5

Code of Conduct:

I P. Hemachandran 1912525440 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence

Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand
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1. Concept Map: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship

Database Management System (DBMS)

- A software system used to create, store, retrieve, update, and manage databases.
- Uses a **Data Model** to organize and represent data.
- A **Data Model** defines how data is structured, stored, and related.
- An **Entity-Relationship (ER) Diagram** is used during database design.
- An **Entity** represents a real-world object such as Student, Employee, or Product.
- An **Attribute** describes the properties of an entity (e.g., Student_ID, Name, Age).
- A **Relationship** shows how two or more entities are connected (e.g., Student enrolls in Course).
- A **Schema** defines the overall logical structure of the database, including tables and relationships.
- An **Instance** represents the actual data stored in the database at a specific point in time.
- The DBMS uses the schema to manage database instances efficiently.

2. Concept Map: ANSI/SPARC Three-Level Architecture

ANSI/SPARC Architecture

- ❖ Divides the database into three levels.
- ❖ **External Level** provides different views for different users.
- ❖ Each user sees only the required part of the database.
- ❖ **Conceptual Level** represents the complete logical structure of the database.
- ❖ It defines entities, relationships, and constraints.
- ❖ **Internal Level** describes the physical storage of data on disks.
- ❖ It includes file organization, indexing, and storage techniques.
- ❖ **Logical Data Independence** allows changes in the conceptual schema without affecting external views.
- ❖ **Physical Data Independence** allows changes in storage methods without affecting the conceptual schema.
- ❖ This architecture improves flexibility, security, and maintainability.

3. Concept Map: File-Based System vs Database System

File-Based System

- ✓ Stores data in separate files.
- ✓ Causes data redundancy because the same data is stored multiple times.
- ✓ Leads to data inconsistency when files are updated separately.
- ✓ Makes data sharing difficult.
- ✓ Provides weak security.
- ✓ Backup and recovery are difficult.
- ✓ Data retrieval is slow and inefficient.

Database Management System

- Stores data in a centralized database.
- Reduces redundancy through normalization.
- Maintains consistency using constraints.
- Allows multiple users to access shared data.
- Provides security through authentication and authorization.
- Supports backup and recovery.
- Improves data integrity and performance.

4. Concept Map: ER Diagram Components

ER Diagram

- Contains **Entities**.
- Includes **Weak Entities**, which depend on strong entities.
- Has different **Attribute Types**:
 - Simple Attribute
 - Composite Attribute
 - Multivalued Attribute
 - Derived Attribute
- Represents **Relationship Types**:
 - One-to-One
 - One-to-Many
 - Many-to-Many
- Uses **Participation Constraints**:
 - Total Participation
 - Partial Participation

5. Concept Map: Hierarchical, Network, and Relational Data Models

Hierarchical Model

- Organizes data in a tree structure.
- Supports one-to-many relationships.

- Suitable for organizational hierarchies.

Network Model

- Organizes data as a graph.
- Supports many-to-many relationships.
- Suitable for interconnected data.

Relational Model

- Stores data in tables.
- Uses rows and columns.
- Supports SQL
 - Uses primary and foreign keys.

6. Concept Map: Enhanced Entity-Relationship (EER) Model

EER Model

- Extends the basic ER model.
- Supports **Generalization**, which combines similar entities into a superclass.
- Supports **Specialization**, which divides a superclass into subclasses.
- Supports **Aggregation**, which treats relationships as higher-level entities.
- Helps model complex real-world applications.

7. Concept Map: DBMS Software Architecture

User

→ Sends a query.

Query Processor

→ Parses and validates the query.

Query Optimizer

→ Chooses the most efficient execution plan.

Execution Engine

→ Executes the optimized query.

Storage Manager

→ Retrieves or updates data.

Database

→ Stores data permanently.

Result

→ Returned to the user.

8. Concept Map: Data Models and Real-World Applications

1. Hierarchical Model

- Used in file systems and organizational structures.

2. Network Model

- Used in social networks and telecommunication systems.

3. Relational Model

- Used in banking systems, hospital management, libraries, colleges, and e-commerce applications.

Each model is suitable based on its structure and relationship capabilities.

9. Concept Map: Schema, Instance, and Data Independence

Schema

- Defines the database structure.
- Changes rarely.

Instance

- Represents the current contents of the database.
- Changes frequently.

Logical Data Independence

- Allows changes in the conceptual schema without affecting user views.

Physical Data Independence

- Allows changes in physical storage without affecting the logical schema.

10. Concept Map: Roles and Responsibilities in a DBMS Environment

Database Administrator (DBA)

- Creates and maintains the database.
- Controls security.
- Performs backup and recovery.
- Monitors performance.

Application Developer

- Designs database applications.
- Writes SQL queries.
- Develops user interfaces.
- Maintains software.

End User

- Accesses the database.
- Inserts, updates, deletes, and retrieves data.

- Uses applications to perform daily tasks.